
Encoding Inductive Biases in Simulation-based inference

Anonymous Author
Anonymous Institution

Abstract

We introduce Amortized Implicit Copula Generative Models (AICGMs), a simulation-based inference method leveraging the expressivity of implicit copulas. This framework provides a novel way to encode many types of inductive biases, and naturally handles parameters with both continuous and discrete support. Preliminary experiments show that AICGMs can accurately recover the posterior distributions with only 10^3 simulator calls.

1 Introduction

Inference in stochastic simulators lies at the heart of many scientific modeling workflows. Parameters of interest are inputs to the simulation, and the simulator returns observations consistent with the practitioner’s understanding of the parameter-data relationship. Due to the likelihood implied by the simulator usually being intractable, Simulation-based inference (SBI) methods are often required to perform inference on the unobserved parameters. SBI methods learn to represent an approximate posterior from prior-simulator data. While some approaches learn parametric likelihoods or posteriors (Papamakarios and Murray (2016); Drovandi et al. (2018)), others build sample-based mechanisms learning to sample an approximate posterior, more akin to generative models (Reuter et al., 2025; Gloeckler et al., 2024; Arruda et al., 2025; Wildberger et al., 2023).

While generative methods, based on Diffusions or Flows, are capable of learning a rich class of distributions, and are often endowed with universal approximation guarantees, they also have two key limitations. Firstly, they do not naturally support the incorporation of inductive biases on properties such as support, multi-modality

skewness and structural dependence on observables. While recent work by Carrasco-Pollo et al. (2026) aims to improve this, most generative methods must learn these properties exclusively from data, which can be scarce if simulations are expensive. Secondly, it is not straight-forward how to handle discrete parameters, especially in mixed settings. If the number and supports of the discrete parameters are small, one may treat this as a Bayesian model selection problem. But, this quickly becomes infeasible as the cardinality of the discrete support grows.

In this work, we present a method that handles both of these limitations naturally, thereby enabling amortized Simulation-based inference with bespoke inductive biases for arbitrary combinations of continuous and discrete parameters. This is achieved by separating marginal posteriors from joint dependence, using an implicit copula framework. The de-coupling allows us to perform Neural-posterior estimation (NPE) per marginal, and then tie marginals together by learning an implicit copula.

We show that this approach is principled theoretically, and demonstrate its effectiveness in a number of experiments.

2 Preliminaries

Notation Over-loading notation, we denote all probability density functions by $p(\cdot)$ and their approximations by $q(\cdot)$, all cumulative distribution functions (CDFs) by $F(\cdot)$, all quantile functions by $F^{-1}(\cdot)$, and all copulas by $\mathcal{C}(\cdot)$, with the argument indicating which random variable the distribution is on. We consider data generating processes (DGPs) of the form $(\theta, x) \sim p(\theta)p(x | \theta)$, with parameters $\theta \in \Theta \subset \mathbb{R}^m$ and observations $x \in \mathcal{X} \subset \mathbb{R}^d$. The only assumption we make for the prior on parameters, $p(\theta)$, and the observation model, $p(x | \theta)$, is that they can be sampled from. All learnable parameters are collected in ϕ .

Copulas Any random vector, $\theta \in \mathbb{R}^m$, with $\theta \sim p(\theta)$ has an associated random vector $u \sim \mathcal{C}(\theta)$ with $\mathcal{C}(\theta)$ the copula distribution of $p(\theta)$ and $u \in [0, 1]^m$ (Sklar, 1959).

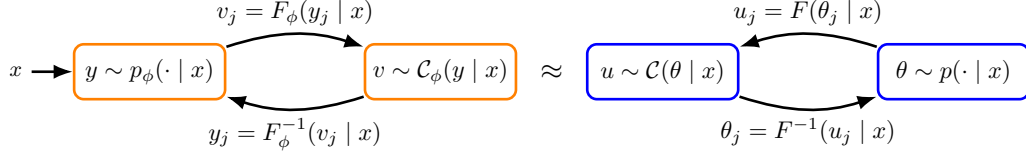


Figure 1: Graphical representation of the amortized implicit copula generative model. The learnable latent model implicitly defines the copula $\mathcal{C}_\phi(y | x)$ (orange), which is matched to the copula of the true posterior (blue).

Denoting the cumulative distribution function (CDF) of θ by $F(\theta)$ and the marginal CDFs by $\{F(\theta_j)\}_{j=1}^m$, by the probability integral transform (PIT) we have for $\theta \sim p(\theta)$

$$u_j = F(\theta_j) \sim \mathcal{U}(0, 1), \quad (1)$$

with

$$u = (F(\theta_j))_{j=1}^m \sim \mathcal{C}(\theta), \quad (2)$$

and also that, if $u \sim \mathcal{C}(\theta)$, then

$$(F^{-1}(u_j))_{j=1}^m \sim p(\theta) \quad (3)$$

where $F^{-1}(\cdot)$ is the quantile function of $F(\cdot)$. It is this mechanism transforming samples from $\mathcal{C}(\theta)$ into samples from $p(\theta)$ that underlies Implicit Copula generative models (ICGMs) (Janke et al., 2021).

Implicit Copula Generative Models (Janke et al., 2021) ICGMs use i.i.d. samples $\{\theta^i\}_{i=1}^n \sim p(\theta)$ to learn a copula distribution, $\mathcal{C}_\phi(\cdot)$, approximating the true copula, $\mathcal{C}(\cdot)$. Given the marginal inverse CDFs, $\{F^{-1}(u_j)\}_{j=1}^m$, this provides a mechanism to approximately sample $p(\theta)$, by first sampling $\mathcal{C}_\phi(\cdot)$ and then applying Equation (3). Since the marginal CDFs are in general unknown, these are also estimated from the respective marginal samples, $\{\theta_j^i\}_{i=1}^n$.

3 Amortizing Implicit Copula Generative Models

Extending the ICGM framework, our proposed method learns an amortized implicit copula, $\mathcal{C}_\phi(\cdot | x)$ from joint samples, $\{(x^i, \theta^i)\}_{i=1}^n$. To this end we first learn the marginal posteriors, $\{p(\theta_j | x)\}_{j=1}^m$, which are then used to train $\mathcal{C}_\phi(\cdot | x)$, capturing the joint dependencies in $p(\theta | x)$. The decomposition into marginals and the copula allows to learn the joint dependencies in $p(\theta)$ independently of the marginals. A graphical representation of our model is given in Figure 1.

Inductive biases via marginal posteriors The ability to model the marginal posteriors separately allows significant flexibility in encoding inductive biases.

Throughout, the basic idea will be to perform Neural-posterior estimation (NPE) with mixture density networks (MDNs) (Papamakarios and Murray, 2016),

$$q_\phi(\theta_j | x) = \sum_{l=1}^L \pi_\phi^l(x) q_\phi^l(\theta_j | x) \quad (4)$$

but the mixture components will be tailored to the respective parameter. The only requirement for these uni-variate mixture-models is that the CDF is tractable, i.e., that each mixture component has a tractable CDF.

For parameters supported on $\mathbb{R}$, an amortized Gaussian-mixture model (GMM) is a good default. If one suspects a heavy-tailed posterior, a Student-T or even a Cauchy component could be added here. For restricted supports such as $[a, \infty)$ or $[a, b]$, with $a < b \in \mathbb{R}$, Gamma, Beta, Uniform or truncated Gaussians may be appropriate. For discrete $\theta_j \in \mathbb{N}$, Poisson components are possible, as well as the more expressive Negative Binomial.

A strength of this approach is that the training procedure is fully independent of the choice of mixture components. In particular, the training objective is

$$\int p(x) \text{KL}(p(\theta_j | x) \| q_\phi(\theta_j | x)) dx \quad (5)$$

$$= -\mathbb{E}_{p(\theta_j, x)} [\log q_\phi(\theta_j | x)] + C, \quad (6)$$

with C constant in ϕ . This objective can be estimated unbiasedly from marginal samples $\{(\theta_j^i, x^i)\}_{i=1}^n$ (see Appendix A). Note that, further inductive biases may be encoded in the training procedure. For instance, multi-modality can be induced and maintained via initialization of mixture components and penalizing component closeness.

At given $x \in \mathcal{X}$, the estimates $F_\phi(\theta_j | x)$ can be inverted, e.g., via bisection, to give approximate quantile functions, $F_\phi^{-1}(u_j | x)$.

Joint dependence The true copula distribution $\mathcal{C}(\theta | x)$ encodes the joint dependencies of the posterior $p(\theta | x)$. In our model, the true copula is approximated by the copula $\mathcal{C}_\phi(y | x)$, implicitly defined as the copula of a latent variable, $y \in \mathbb{R}^m$. In particular,

Algorithm 1 Sampling $q_\phi(\theta | x)$

Require: observation x , trained MDN $p_\phi(y | x)$ with marginal CDFs $F_\phi(y_j | x)$ and approximate posterior quantile functions $F_\phi^{-1}(u_j | x)$ for $j = 1, \dots, m$

- 1: sample $y \sim p_\phi(\cdot | x)$
- 2: set $v = (F_\phi(y_j | x))_{j=1}^m$ (Equation (2))
- 3: set $\theta = (F_\phi^{-1}(v_j | x))_{j=1}^m$ (Equation (3))
- 4: **return** θ

$\mathcal{C}_\phi(y | x)$ is the copula associated with the conditional distribution over latents $p_\phi(y | x)$. The distribution over latents needs to be flexible enough to match the dependencies of the true posterior, but also provide tractable marginals, such that Equation (2) can be used to sample its copula, $\mathcal{C}_\phi(y | x)$. The example considered throughout this work is a Gaussian MDN,

$$p_\phi(y | x) = \sum_{k=1}^K w_\phi^k(x) p_\phi^k(y | x), \quad (7)$$

with

$$p_\phi^k(y | x) = \mathcal{N}(y | \mu_\phi^k(x), \Sigma_\phi^k(x)), \quad (8)$$

parameterized conditionally on x by an appropriate neural network. Note that if a richer class of distributions is required, this Gaussian MDN can be upgraded to a probabilistic circuit, while maintaining tractable marginals (Choi et al., 2020).

The implicit copula may be sampled from using Equation (2) and the marginal CDFs, $\{F_\phi(y_j | x)\}_{j=1}^m$, as

$$y \sim p_\phi(y | x), \quad v = (F_\phi(y_j | x))_{j=1}^m \sim \mathcal{C}_\phi(y | x). \quad (9)$$

This is theoretically sound also in the conditional setting, following a simple extension of the PIT, presented in Proposition 1.

Proposition 1 (Conditional PIT). *Let $(x, y) \sim p(x, y)$, then*

$$\mathbb{P}_{x,y}(F(y | x) \leq u) = u, \text{ i.e., } F(y | x) \sim \mathcal{U}(0, 1). \quad (10)$$

Proof. $\mathbb{P}_{x,y}(F(y | x) \leq u) = \mathbb{P}_x \mathbb{P}_{y|x}(F(y | x) \leq u) = u$ $\square$

For a trained generator network we then have $\mathcal{C}_\phi(y | x) \approx \mathcal{C}(\theta | x) \quad \forall x \in \mathcal{X}$ (see Figure 1). For any $x \in \mathcal{X}$, the learned approximate amortized posterior, $q_\phi(\theta | x)$, can then be sampled as shown in Algorithm 1.

Randomized PIT for discrete parameters In order to extend implicit copulas to discrete parameters,

we employ the randomized PIT. In particular, following Rüschendorf (2009), defining for a parameter $\theta_j \in \mathbb{N}$

$$u = F(\theta_j - 1 | x) + \alpha(F(\theta_j | x) - F(\theta_j - 1 | x)), \quad (11)$$

and $\alpha \sim \mathcal{U}(0, 1)$, it holds that $u \sim \mathcal{U}(0, 1)$.

By Proposition 1, this holds again for any fixed x . The random mapping to the unit interval in Equation (11) can be viewed as the discrete analogue to Equation (1), and allows for a principled way to incorporate discrete parameters into our model. Mapping from unit-space back into parameter space is achieved by

$$\theta_j = F_\phi^{-1}(v_j | x) = \min\{t \in \mathbb{N} : F_\phi(t | x) \geq v_j\}, \quad (12)$$

in line 3 of Algorithm 1 for discrete parameters.

Amortized training Since we only have access to both the true posterior copula, $\mathcal{C}(\theta | x)$ and the learnable $\mathcal{C}_\phi(y | x)$, via samples, a purely sample-based training objective is required. For fixed $x \in \mathcal{X}$ and following Janke et al. (2021), we propose to minimize the conditional squared energy distance (Székely et al., 2004), $ED^2(\mathcal{C}(\theta | x) \| \mathcal{C}_\phi(y | x))$. Proposition 1 then allows to amortize this over x , and we minimize

$$\mathbb{E}_{p(x)}[ED^2(\mathcal{C}(\theta | x) \| \mathcal{C}_\phi(y | x))] \quad (13)$$

$$= \mathbb{E}_{p(x)} 2\mathbb{E}_{u \sim \mathcal{C}(\theta | x), v \sim \mathcal{C}_\phi(y | x)}[\|u - v(\phi)\|_2] \quad (14)$$

$$- \mathbb{E}_{p(x)} \mathbb{E}_{v, v' \sim \mathcal{C}_\phi(y | x)}[\|v(\phi) - v'(\phi)\|_2] + C \quad (15)$$

with respect to parameters ϕ .

Given a set of joint samples $\{(x^i, \theta^i)\}_{i=1}^N$ from $p(\theta)p(x | \theta)$, and the ability to sample latents from $p_\phi(y | x)$, this training objective can be estimated unbiasedly. This is because samples from the two copulas can simply be obtained via Equations (1), (2) and (11) from the joint and latent samples. Then, any gradient-based optimization routine may be employed to minimize it. See Appendix B for more detail on the estimation of the objective and training specifics.

4 Empirical evaluation

To illustrate the mechanics of our model, we visualize the two-step posterior sampling procedure in Figure 2. The DGP learned here is

$$(\theta, x) \sim \mathcal{N}(\theta | 0, \Sigma_0) \mathcal{N}(x | \theta, \Sigma_1),$$

a conjugate Gaussian model. It can be seen that the distribution over latents, $p_\phi(y | x)$ has learned the negative correlation of the first two components, while the margins are on vastly different scales. This is by design, since marginal information only enters in the second step, where the models $q_\phi(\theta_j | x)$ are applied.

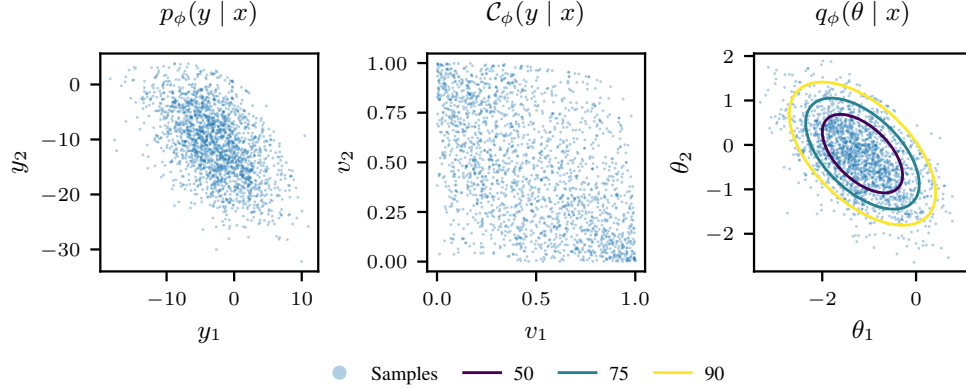


Figure 2: Two step sampling with the implicit copula framework. *Left*: Samples from the distribution over latents, $p_\phi(y | x)$. *Middle*: Samples from the learned implicit copula, $C_\phi(y | x)$ obtained via Equation (2). *Right*: Samples from the approximate posterior, obtained by applying Equation (3). Shown are the first two dimensions of a 10d non-isotropic Normal-Normal conjugate DGP.

Three further experiments are presented, each showcasing distinct features of our model. Performance is reported in terms of C2ST (Lueckmann et al., 2021), in Table 1. We implement our method in JAX (Bradbury et al., 2018) and make our code available at <https://anonymous.4open.science/r/IC-SBI-D3F1>.

Encoding structure The DGP samples

$$(\theta, x) \sim \mathcal{N}(\theta | 0, \alpha I) \mathcal{N}(x | \theta, \sigma^2 I),$$

a conjugate Gaussian model. The true posterior is also an isotropic Gaussian, but, more importantly, the marginal posterior variances, $\mathbb{V}[\theta_j | x]$, are actually all equal and independent of x . This knowledge can of course easily be encoded in the network architecture of the MDNs, $q_\phi(\theta_j | x)$. Table 1 shows that while 10^2 joint samples are not sufficient to learn the posterior, 10^3 already give a decent approximation, and 10^4 make the learned posterior effectively indistinguishable from the true one.

Encoding bounded support $\mathbb{R}, [a, \infty)$ and $[a, b]$ The DGP samples the prior

$$p(\theta) = \mathcal{N}(\theta_1 | 0, 1) \text{Gamma}(\theta_2 | 1, \theta_1^2) \text{Beta}(\theta_3 | 2, \theta_2),$$

simulates $c \sim \text{Bin}(20, \theta_3)$, $z \sim \mathcal{N}(\theta_1, \theta_2)$, and returns $x = c + z$. It can be seen that $\theta_1 \in \mathbb{R}$, $\theta_2 \in [0, \infty)$ and $\theta_3 \in [0, 1]$. Further, dependence among parameters is introduced in the prior, which carries forward to the intractable posterior. Our model handles these mixed supports out-of-the-box via appropriate marginal MDNs, $q_\phi(\theta_j | x)$. In particular, natural choices for mixture component families in this case are Gaussian, Gamma and Beta, for θ_1, θ_2 and θ_3 , respectively. It can be seen from Table 1 that already 10^3 samples allow for a very accurate approximate posterior.

Table 1: C2ST scores averaged over 20 draws $x \sim p(x)$.

DGP \ Samples	10^2	10^3	10^4	10^5
Gaussian	0.89	0.63	0.51	0.50
Bounded Support	0.67	0.53	0.51	0.50
Cont. + Discrete	0.63	0.57	0.54	0.53

Handling discrete parameters This experiment introduces a discrete parameter with countably infinite support, making tools such as Bayesian model selection infeasible. The DGP samples the prior

$$p(\theta) = \text{Beta}(\theta_1, 2, 3) \text{NegBin}(\theta_2 | 10, \theta_1) \text{Gamma}(\theta_3 | 2, \theta_2),$$

simulates $c \sim \text{Bin}(\theta_2, \theta_1)$, $z \sim \mathcal{N}(0, \theta_3)$, and returns $x = c + z$. As before, dependence is introduced in the prior, and the true posterior is intractable. On the marginal side, we choose mixture component families to be Beta, Negative Binomial and Gamma, for θ_1, θ_2 and θ_3 , respectively. Integrating the discrete-valued θ_2 into the training of the implicit copula is as simple as mapping samples $\{\theta_2^i\}_{i=1}^n$ into unit space via Equation (11). Table 1 shows how working in copula space is an effective way to handle both continuous and discrete parameters simultaneously.

5 Discussion and future work

We presented AICGMs, an amortized variant of the Implicit Copula framework capable of handling both continuous and discrete parameters. In future work we will extend the empirical evaluation of our method to higher dimensions and real-world problems. Additionally, other losses than the Energy distance could be explored.

References

- Jonas Arruda, Niels Bracher, Ullrich Köthe, Jan Hase-nauer, and Stefan T Radev. Diffusion models in simulation-based inference: A tutorial review. *arXiv preprint arXiv:2512.20685*, 2025.
- James Bradbury, Roy Frostig, Peter Hawkins, Matthew James Johnson, Chris Leary, Dougal Maclaurin, George Necula, Adam Paszke, Jake VanderPlas, Skye Wanderman-Milne, and Qiao Zhang. JAX: composable transformations of Python+NumPy programs, 2018. URL <http://github.com/jax-ml/jax>.
- Jorge Carrasco-Pollo, Floor Eijkelboom, and Jan-Willem van de Meent. Pawstterior: Variational flow matching for structured simulation-based inference. *arXiv preprint arXiv:2602.13813*, 2026.
- Y Choi, Antonio Vergari, and Guy Van den Broeck. Probabilistic circuits: A unifying framework for tractable probabilistic models. *UCLA*. URL: <http://starai.cs.ucla.edu/papers/ProbCirc20.pdf>, 6, 2020.
- Christopher C Drovandi, Clara Grazian, Kerrie Mengersen, and Christian Robert. Approximating the likelihood in abc. *Handbook of approximate bayesian computation*, pages 321–368, 2018.
- Manuel Gloeckler, Michael Deistler, Christian Weilbach, Frank Wood, and Jakob H Macke. All-in-one simulation-based inference. *arXiv preprint arXiv:2404.09636*, 2024.
- Tim Janke, Mohamed Ghanmi, and Florian Steinke. Implicit generative copulas. *Advances in Neural Information Processing Systems*, 34:26028–26039, 2021.
- Jan-Matthis Lueckmann, Jan Boelts, David Greenberg, Pedro Goncalves, and Jakob Macke. Benchmarking simulation-based inference. In *International conference on artificial intelligence and statistics*, pages 343–351. PMLR, 2021.
- George Papamakarios and Iain Murray. Fast ϵ -free inference of simulation models with bayesian conditional density estimation. *Advances in neural information processing systems*, 29, 2016.
- Arik Reuter, Tim GJ Rudner, Vincent Fortuin, and David Rügamer. Can transformers learn full bayesian inference in context? *Forty-second International Conference on Machine Learning.*, 2025.
- Ludger Rüschendorf. On the distributional transform, sklar’s theorem, and the empirical copula process. *Journal of statistical planning and inference*, 139(11): 3921–3927, 2009.
- M Sklar. Fonctions de répartition à n dimensions et leurs marges. In *Annales de l’ISUP*, volume 8, pages 229–231, 1959.
- Gábor J Székely, Maria L Rizzo, et al. Testing for equal distributions in high dimension. *InterStat*, 5(16.10): 1249–1272, 2004.
- Jonas Wildberger, Maximilian Dax, Simon Buchholz, Stephen Green, Jakob H Macke, and Bernhard Schölkopf. Flow matching for scalable simulation-based inference. *Advances in Neural Information Processing Systems*, 36:16837–16864, 2023.

A Training the marginal MDNs

For the marginal posteriors a MDN can be fit purely from samples $\{(\theta_j^i, x^i)\}_{i=1}^n$ by minimizing the following objective.

$$\int p(x) \text{KL}(p(\theta_j | x) \| q_\phi(\theta_j | x)) dx \quad (16)$$

$$= -\mathbb{E}_{p(\theta_j, x)} [\log q_\phi(\theta_j | x)] + C, \quad (17)$$

which we can estimate in an unbiased manner via

$$\mathbb{E}_{p(\theta_j, x)} [\log q_\phi(\theta | x)] \approx \frac{1}{n} \sum_i \log q_\phi(\theta_j^i | x^i). \quad (18)$$

B Training the implicit copula

The copula implied by the distribution over latents, $p_\phi(y | x)$ is learn by minimizing the energy distance between the observed copula samples $u \in [0, 1]^d$ and pseudo-copula samples obtained from $p_\phi(y | x)$. We write

$$\begin{aligned} & \text{ED}^2(\mathcal{C}(\theta | x) \| \mathcal{C}_\phi(y | x)) \\ &= 2\mathbb{E}_{\theta \sim p(\theta | x), y \sim p_\phi(y | x)} [\|u(\theta, x) - v(y, x; \phi)\|] - \mathbb{E}_{y, y' \sim p_\phi(\cdot | x)} [\|v(y, x; \phi) - v(y', x; \phi)\|] + C \end{aligned} \quad (19)$$

where v denotes samples mapped to the copula space via the marginal CDFs $p_\phi(y | x)$. Approximate samples from the true posterior copula, $\mathcal{C}(\theta | x)$ are obtained by component-wise mapping joint samples (θ^i, x^i) to unit-space via the learned marginal posteriors, $q_\phi(\theta_j | x)$, as suggested by Equation (2).

Expectations with respect to the GMM are estimated via Monte Carlo using the Gumbel-Softmax straight-through estimator for differentiable component selection. Specifically, at each training step we:

1. Draw L samples $y^{(l)}$ from $p_\phi(y | x)$ using Gumbel-Softmax to select components. The forward pass uses hard one-hot assignments while gradients flow through the continuous softmax relaxation.
2. Map each $y^{(l)}$ to the copula space via the marginal CDFs to obtain $v^{(l)}$.
3. Estimate the energy distance using

$$\widehat{\text{ED}}^2 = \frac{1}{B} \sum_{b=1}^B \frac{2}{L} \sum_{l=1}^L \|u(\theta^b, x^b) - v(y^l, x^b; \phi)\| - \frac{2}{L(L-1)} \sum_{l < l'} \|v(y^l, x^b; \phi) - v(y^{l'}, x^b; \phi)\| + C \quad (20)$$

Parameters are optimised with AdamW using default hyperparameters.